# RAJ TEJPAL KHATIK

*AI/ML Engineer | Agentic AI Systems | GenAI Research*

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Applied AI engineer who has completed the MSc in Applied Artificial Intelligence at the University of Warwick (WMG; programme completed September 2026, final result pending), with 2+ years of prior industry experience in data analytics and ML engineering. I design multi-agent, retrieval-augmented systems — from causal-chain equity analysis to hallucination detection — and publish the research behind each one. Five systems deployed and one open-source agent, four papers under review, dissertation submitted. Open to AI/ML Engineer, GenAI Engineer, and Agentic AI roles, and exploring PhD positions in CS/AI and applied economics.

## TECHNICAL SKILLS

**Programming & Core:** Python, Data Structures, Algorithms, Probability & Statistics, SQL/SQLite, Git/GitHub version control

**LLMs & Agentic AI:** Claude (Anthropic), GPT-4o / GPT-4o-mini, Gemini, LangGraph, LangChain, MCP (Model Context Protocol) tool gateways, multi-agent orchestration, tool calling / function calling, structured outputs, prompt engineering

**RAG & Retrieval:** Retrieval-Augmented Generation, FAISS vector database, embeddings, semantic search

**Evaluation & Observability:** LangSmith tracing, model/agent evaluation pipelines, statistical testing (McNemar's, Cochran's Q, logistic regression, chi-square), guardrails, adversarial red-teaming, human-in-the-loop controls

**Data & Analytics:** NumPy, Pandas, data pipeline design, ETL, Power BI, structured data extraction

**APIs & Deployment:** REST API design & consumption, OAuth 2.0, secure credential handling, FastAPI, Streamlit Community Cloud, Cloudflare Pages/Workers, automated adversarial/QA testing, technical documentation

## WORK EXPERIENCE

**H&B Infotech — AI Engineer** **MAR 2024 - MAR 2025**

- Built ML pipelines (Python, NumPy, Pandas) to prepare and process data for model training, integrated into existing backend systems — improving analysis accuracy and reporting efficiency.

**H&B Infotech — Software Engineer** **MAR 2023 - MAR 2024**

- Engineered backend systems and data pipelines (Python, MySQL, Pandas, NumPy) powering internal reporting tools, with data validation and front-end integration work.

**H&B Infotech — Software Engineer Intern** **AUG 2022 - FEB 2023**

- Developed and validated Python-MySQL modules to support internal systems; built front-end interfaces (HTML, CSS, JavaScript, React) for internal tools.

**Fusion UX Lab, University of Warwick — Play Tester** **NOV 2025 - AUG 2026**

- Structured playtesting of games and interactive systems; wrote usability and player-experience feedback.

## AI / ML PROJECTS

Six AI/ML systems (five deployed, one open-source), each built on a shared architecture (LangGraph orchestration, FAISS retrieval, Streamlit interfaces) applied to a different domain. Full write-ups: portfolio-raj.pages.dev/#projects

**SemiBot — Multi-Agent Causal-Chain RAG System (Flagship)** **2026**

- 18-factor causal-chain RAG system tracing macro conditions to sector outcomes for semiconductor equities, with per-factor attribution rather than a single opaque verdict. LangGraph, FAISS, GPT-4o-mini, MCP/Tavily tool-gateway integration, SQLite checkpointing, LangSmith observability.
- Backtested across 48 tickers x 10 dates (426 observations): accuracy scales from 84.0% (>$1T market cap) to 41.7% (mid-cap), confirmed via logistic regression ($p<0.001$) and robust to sub-sector controls.

Demo | Code | SSRN — under review (ESWA)

**GoldBot — 9-Factor Gold Safe-Haven Analyser** **2026**

- Cross-regime backtesting (4 structurally distinct historical regimes) built specifically to separate genuine event discrimination from lucky trend alignment.
- Diagnostic finding: across 18 distinct events, Strength% output stayed within a 37-50% band (SD=4.2) — exposing trend-matching rather than true discrimination, a failure mode a single backtest would have missed.

Demo | Code | SSRN — under review

**SilverBot — Dual-Regime Industrial/Monetary Analyser** **2026**

- Decomposes silver's price behaviour into independent Industrial and Monetary strength scores, addressing a documented gap in the classical gold/silver duality literature.
- Industrial signals beat Monetary in 3 of 4 regimes (61.7% vs 40.0%, $p=0.041$ in the largest sample) — but the pattern cleanly reverses during the 2020 liquidity crisis (Monetary 95.0% vs Industrial 65.0%), refining the classical Batten-Ciner-Lucey framework.

Demo | Code | SSRN — under review

**Hallucination Detector — LLM Reliability Ensemble (Flagship)** **2026**

- Six-signal ensemble (NLI, retrieval grounding, three-model LLM-judge panel, cross-sample self-consistency) as a parallel LangGraph pipeline, evaluated with formal significance testing rather than accuracy alone.
- 72.4-92.2% accuracy across three benchmarks. Quadrant analysis overturned the system's own design assumption: "flagged + consistent" claims were 100% real hallucinations (n=4, small sample), beating the architecture's own "high-confidence" category (66.7%).

Demo | Code | SSRN — under review

**HITL Email Agent — Human-in-the-Loop LangGraph Agent (Repo only)** **202** **6**

- Human-in-the-loop email drafting agent that never sends without approval. Binary Response/Ignore triage stage, full multi-turn interrupt/resume via LangGraph, dual Gmail OAuth, and a Manifest V3 Chrome extension for inline drafting alongside a standalone Streamlit interface.
- Credentials kept out of the repository (OAuth files git-ignored). Two interfaces (v1 Streamlit, v2 Chrome extension with version-history tabs) shipped and maintained side by side.

Code — open-source

**Retail Research Assistant — Full S&P 500 Research Platform** **2026**

- Full-coverage discovery, research, comparison, and portfolio analysis across all 503 S&P 500 constituents, spanning six pages from dashboard to guided learning. Governing principle: the LLM narrates what the data shows — it never tells the user what to buy.
- Passed a 100-question adversarial safety test, 100/100, results committed to the repo. Implements 9 of 10 recognized agentic-AI design patterns, with the one omission (MCP) explicitly justified rather than silently skipped.

Demo | Code

## PUBLICATIONS

- **A Domain-Specific Causal Chain RAG System for Semiconductor Equity Analysis** — SemiBot. SSRN 7199998, under review. Presented at University of Warwick.
- **GoldBot: A Multi-Factor Safe-Haven Analyzer for Gold** — Cross-Regime Evidence for Trend-Matching Rather Than Event Discrimination. SSRN 7200198, under review.
- **SilverBot: A Dual-Regime Analyzer for Silver's Industrial-Monetary Duality** — Cross-Regime Evidence for a Crisis-Type-Dependent Reversal. SSRN 7203201, under review.
- **Multi-Method Ensemble Framework for LLM Hallucination Detection.** SSRN 7200222, under review. Presented at University of Warwick.

## DISSERTATION

**Understanding the Role of Artificial Intelligence in Life Cycle Assessment** — supervised by Dr. You Wu, WMG, University of Warwick. Submitted September 2026; result pending.

Dashboard | Code

- **Obj. 1 — Landscape Review:** Systematic literature review (538 papers screened from 1,509 candidates, 209 analysed in full) plus a structured review of 8 commercial AI-powered LCA platforms against the four ISO 14040/14044 stages. Finding: none of the 8 platforms target AI-assisted goal-and-scope drafting — the gap Objective 3 addresses.
- **Obj. 2 — Applied ML on LCI Data:** Multi-method unsupervised learning on ecoinvent 3.11 (25,412 records). Reduced 68 redundant climate indicators to 7 distinct features; cross-validated 3 clustering algorithms and 2 anomaly detectors. KMeans/Agglomerative converged (silhouette 0.689/0.686); Isolation Forest flagged 509 anomalies (2%), traced to a genuine data artefact.
- **Obj. 3 — LLM Evaluation on LCA Tasks:** Evaluated 6 LLMs (Claude Haiku 4.5, GPT-4o-mini, GPT-5.6 Terra, Gemini 3.5 Flash, Llama 3.3 70B, GPT-OSS 120B) on structured extraction and ISO 14044 drafting. A harder, paraphrased variant dropped accuracy by up to 65 points (GPT-4o-mini 91%->26%); Gemini 3.5 Flash held up best (100%->86%), significantly outperforming the two weakest models (Cochran's $Q=66.7$, $p<0.000001$).

## EDUCATION

**University of Warwick, WMG — MSc Applied Artificial Intelligence** **SEP 2025 - SEP 2026**

Programme completed; final result pending.

**Gujarat Technological University — BE, Computer Science and Engineering** **AUG 2021 - MAY 2025**

**CGPA:** 8.49/10 (First Class with Distinction)

## CERTIFICATIONS

- **Financial Markets** — Coursera, Yale University, taught by Prof. Robert Shiller
- **GenAI Enhanced Financial Analysis Specialization** — Coursera, Microsoft (Generative AI, Excel Copilot & Power BI for financial forecasting)
- **Generative AI Engineering** — Coursera, IBM

*References available on request*

*Right to work: UK Graduate visa application submitted — no sponsorship required*